

University of Westminster

School of Electronics and Computer Science

SUBJECT TO EXTERNAL EXAMINER APPROVAL

7BUIS025W Web and Social Media Analytics (2024/25)	
Module leader	Dr Philip Worrall
Units	Individual Coursework
Weighting:	60%
Qualifying mark	40%
Description	Students will be assessed on their ability to develop and test an analytical solution using appropriate theory, tools and techniques.
Covered Learning Outcomes	LO4 - Collect and prepare a social media data set for analytical processing, including user of social media APIs, data cleansing, mapping, and pre-processing. LO5 - Construct and critique a semantic model of a social media conversation for the purpose of assessing consumer opinion to inform product and or marketing decisions.
Handed Out:	Thursday 27th March 2025
Due Dates	1pm Thursday 8th May 2025
Expected deliverables	• Report (in either PDF or DOCX format) • Video Presentation
Method of Submission:	Online via Blackboard.
Type of Feedback and Due Date:	Written feedback on the submission will be provided within 3 weeks after the submission (the mark and comments via BB Rubric). All marks remain provisional until formally agreed by an Assessment Board.

Assessment regulations

Refer to Part 3: Assessment regulations for taught courses of the guide for postgraduate students for a clarification of how you are assessed, penalties and late submissions, what constitutes plagiarism etc.

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark, as a penalty for late submission, except for work which obtains a mark in the range 50 – 59%, in which case the mark will be capped at the pass mark (50%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognised that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Campus Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regarding University Assessment Regulations, please refer to the following website: <https://www.westminster.ac.uk/current-students/guides-and-policies/academic-matters/academic-regulations>.

INDIVIDUAL COURSEWORK

OVERVIEW

In this coursework assignment, students will assume the role of a data scientist working at a social media company.

Students will apply the tools, techniques and relevant theory covered on the module to carry out an investigation into the interests and opinions of an existing online community. The outcome of your investigation will be a report detailing your analysis and a 5-minute video presentation of your findings.

To ensure successful completion of the coursework, students will need to demonstrate their ability to follow the social media modelling process and derive relevant insights.

This coursework **MUST** be completed **INDIVIDUALLY**.

YOUR TASK

You have begun working for a social media company. The social media company's website hosts thousands of large and diverse online communities across a broad range of topics.

The social media company would like to increase its understanding of each community's interests, sentiments and opinions across the topics being discussed. Any support you can provide will help inform the company's marketing strategy.

Select an online community to investigate. This community might be on reddit, YouTube or another social media site for which data is publicly accessible. Collect and prepare a social media dataset using data from the selected community. Carry out an analysis of this dataset using appropriate social media modelling tools and techniques.

You will need to submit a report containing your findings and results. The report should also carefully document the process you have followed, including annotations of any code you have developed or visualisations and figures you have created.

You also need to produce a 5-minute video presentation intended for senior executives of the company. The video presentation will need to focus on providing a high-level overview of the work you have undertaken, highlighting key findings and insights you think are most relevant.

Students are **NOT** permitted to work on the same online community. You **MUST** confirm your choice with the module leader before undertaking the task.

You should refer to the attached marking scheme for a breakdown of how marks are awarded.

DELIVERABLES

The following deliverable(s) should be submitted via the modules Blackboard site (submission links are available under section "Assessment") by **1pm Thursday 8th May 2025**.

1. Report

Suggested word count: 3000 words

Single Microsoft Word File or PDF

The report should document the analysis and modelling you have conducted. The report **SHOULD** consist of the following 5 sections.

1. **INTRODUCTION** – Give an overview of the online community selected for your analysis. Include basic details on the community's size, any relevant history or notoriety, reach and viewership, number of moderators, and formation date etc.
2. **DATA** – Describe the process used to collect and prepare a social media dataset. Your dataset should include comments or posts from the online community selected for analysis. You should illustrate your answer with examples of annotated code that you developed. Apply any pre-processing steps you believe are relevant and prepare your data for further modelling. You are not required to upload the entire dataset, instead you should include a screenshot of the raw data and following the pre-processing stage.
3. **EXPLORATORY** – Carry out an exploratory analysis of your data. Identify popular words, phrases, and associations. Your exploratory analysis should include tables, charts and figures where appropriate to help visualise and describe your data.
4. **TEXT MINING** – Utilise appropriate text-mining models to examine your data, identifying key discussion themes, opinions and sentiments. Support your analysis with tables, charts, and figures to communicate your results and findings
5. **CONCLUSION** – Summarise your key findings and critically evaluate the analysis you have conducted. Provide an evaluation of the current health and status of the online community you have chosen.

2. Video Presentation

Duration: 5 minutes

In the video you should focus on providing a high-level overview of the work that has been undertaken, highlighting key findings and insights you think are most relevant. You should refer to graphs, charts or figures contained in your submitted report. Your face and voice **MUST** be clear in the recording. The duration should not exceed 5 minutes. Videos that exceed 5 minutes in duration will be awarded a mark based only on the first 5 minutes. To create your video presentation, you are advised to use a MS Teams meeting or other suitable desktop recording software (e.g. OBS Studio).

7BUI025W – WEB AND SOCIAL MEDIA ANALYTICS – INDIVIDUAL COURSEWORK - MARKING SCHEME

NOTE: Marks will be allocated in relation to the correctness, completeness, and the quality of the answer provided.

Criteria	Mark Per Component	Mark Provided	Comments
COMPONENT	100 marks		
INTRODUCTION			
Give an overview of the online community selected for your analysis. Include basic details on the community's size, any relevant history or notoriety, reach and viewership, number of moderators, and formation date etc.			
• Clear and concise overview of the community selected for analysis (2 marks) • The extent to which the community is well-researched including details of any relevant history of the community or notoriety (2 marks) • The extent to which quantitative details/summary statistics on the community's size, reach and viewership, number of moderators, and formation date etc are provided. (4 marks)	8		
DATA			
Describe the process used to collect and prepare a social media dataset. Your dataset should include comments or posts from the online community selected for analysis. You should illustrate your answer with examples of annotated code that you developed. Apply any pre-processing steps you believe are relevant and prepare your data for further modelling. You are not required to upload			

the entire dataset, instead you should include a screenshot of the raw data and following the pre-processing stage.			
- Details of keywords, specific URLs or search terms used to gather data (2 marks) - The extent to which the data collection procedure is carefully documented, justified and explained; for example, by including annotations of code developed that sources data from public APIs or endpoints (4 marks) - Evidence that a set of (2-3) relevant pre-processing steps have been correctly applied, for example using before and after screenshots. (8 marks) - The extent which the student demonstrates and understanding of the importance and role of pre-processing in the social media modelling process. (2 mark)	16		
EXPLORATORY			
Carry out an exploratory analysis of your data. Identify popular words, phrases, and associations. Your exploratory analysis should include tables, charts and figures where appropriate to help visualise and describe your data.			
- The extent to which the exploratory analysis is carefully documented, justified and explained; for example, by including annotations of code developed that sources data from public APIs or endpoints (4 marks) - Analysis of popular words and phrases (2 marks) - Examination of word associations/correlations (2 marks) - Time-series analysis of comments/posts (2 marks) - The extent to which results are carefully and correctly interpreted (2 marks) - Clarity and completeness of visualisations used to summarise findings (2 marks)	14		

TEXT MINING			
Utilise appropriate text-mining models to examine your data, identifying key discussion themes and sentiments. Support your analysis with tables, charts, and figures to visualise your results and findings			
- The extent to which the text-mining procedures are carefully documented and explained; for example, by including annotations of code developed that sources data from public APIs or endpoints (4 marks) - Quality of the developed code to model discussion themes and topics (6 marks) - Quality of the developed code use to model sentiments and opinions (6 marks) - Evaluation of goodness of fit and other diagnostic indicators (4 marks) - Extent to which the proposed modelling approach contains elements of creativity, rigour and thoroughness (2 marks) - Extent to which the results and findings are supported by appropriate visualisations, including tables and charts, that are interpreted correctly (6 marks)	28		
CONCLUSION			
Summarise your key findings and critically evaluate the analysis you have conducted. Provide an evaluation of the current health and status of the online community you have chosen.			
- Clear and concise communication of key results (4 marks) - Quality of the critical evaluation of the adopted approach (4 marks) - Intuitive explanation of the results in the context of the original investigation (2 marks) - Identification of potential actions and avenues for extensions of the work (2 marks)	12		

WRITING AND COMMUNICATION			
- The extent to which the report is appropriately formatted, including labelling of figures and tables, referenced and structured (4 marks) - The extent to language used is concise and precise without unnecessary complexity. Language suitable for the intended audience (4 marks)	8		
VIDEO PRESENTATION			
- Clear summary of work conducted and key methodological steps (4 marks) - Key insights highlighted (4 marks) - References to results, figures or table in the main body of the report (2 marks) - Appropriate presentation technique (4 marks)	14		
END			